

3-2 三角函数数值微分

$$f(x) = A \sin(x + B) + C$$

$$f'(x) = A \cos(x + B)$$

$$\frac{f(x + \delta) - f(x - \delta)}{\tau} = A \cdot \frac{\sin(x + \delta + B) - \sin(x - \delta + B)}{\tau} = \frac{2A}{\tau} \cdot \sin(\delta) \cos(x + B)$$

若:

$$f'(x) = \frac{f(x + \delta) - f(x - \delta)}{\tau}$$

则有:

$$\tau = 2 \sin(\delta)$$